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Microvibration Model Development and Validation of a Cantilevered Reaction Wheel Assembly

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Abstract. Reaction wheel assemblies are one of the most important microvibration sources on typical modern satellites. In this paper microvibrations induced by a cantilevered reaction wheel assembly are modelled and validated against microvibration test results. The disturbance model is developed using energy method. A microvibration measurement platform is designed to measure its disturbances. Disturbance test results are analyzed in detail. The peculiar dynamic characteristics such as nonlinearity and high damping of harmonic responses in the test results are discussed. Estimations of damping values used in the disturbance model are introduced. A new method developed to model harmonic excitations is discussed. Furthermore, novel methods to identify harmonics and extract model parameters from test results are presented. The empirical modeling method developed for broadband noise excitations are also introduced and validated.

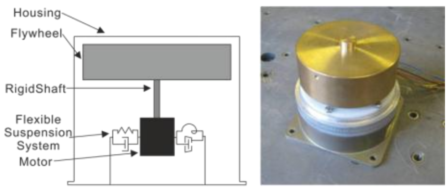
Introduction

In recent years, microvibrations (typically characterized as disturbances with amplitudes in micro-g from several Hz to 1k Hz) are becoming important for satellites with accurately targeted instruments such as SOLAR-B, SDO, JWST, etc. These satellites are usually equipped with vibration-sensitive instruments, resulting in strict requirements for satellite structure stability. Among the various disturbance sources on satellite, reaction wheel assemblies (RWAs) are considered as one of the largest disturbance sources. Due to its complex dynamics, RWA microvibration analysis is often difficult to perform. With the advent of new (and often more advanced) RWA designs, traditional analysis techniques are not effective anymore. Meanwhile, efficiency and accuracy of the current methods need to be improved and new methods need to be developed for new designs. On the other hand, due to ever increasing structure stability requirements, microvibrations interested are extending to higher and broadband frequencies with even lower amplitudes. The modeling of these high frequency disturbances requires important higher harmonics being identified and modelled accurately. The traditionally ignored broadband noise excitations also need to be included in the disturbance model.

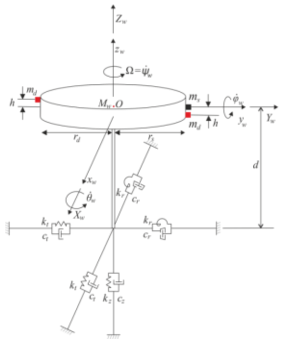
In this paper, the disturbance model of a newly designed cantilevered RWA developed in the past is validated through microvibration tests. Tests are presented including the peculiar nonlinearity and high damping observed. Estimations of damping values from sine-sweep tests are introduced. Broadband noise excitations are modelled using empirical method utilizing bandstop filters. Furthermore methods to identify harmonics and simulate higher harmonic inputs are introduced. The validated disturbance model is extended with broadband noise and higher harmonic excitations to complete disturbance analysis. The modeling methods and programs developed in this paper can also be used for disturbance analysis of conventional RWAs. Note that some commercially sensitive information has been omitted.

RWA Disturbance Model

In this study, a cantilevered RWA is used for disturbance analysis, see Fig. 1 a). The RWA was designed in partnership with SSBV UK. It has common components as a traditional RWA but a soft suspension system instead of rigid ball bearings.



a) Cantilevered RWA under Test



b) RWA Model

Figure 1. Cantilevered RWA

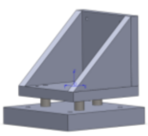
The RWA disturbance model was developed in [1]. The RWA is simplified as the model shown in Fig. 1 b). The three global axes are X_w, Y_w, Z_w; the three body axes are x_w, y_w, z_w; the flywheel mass is M_w; I_r and I_z are radial and polar moment inertia d is the distance from flywheel Centre of Mass (CoM) to the suspension system. "w" indicates RWA only. Suspension system is modelled as a combination of linear and torsional springs and dampers, k_t and c_t are stiffness and damping of the linear spring; k_r and c_r are stiffness and damping of the torsional spring. The model is derived using energy. The final Equations of Motion (EoMs) are presented as following:

$$\begin{bmatrix} M_w & 0 & 0 & 0 & 0 \\ 0 & I_r & 0 & 0 & 0 \\ 0 & 0 & M_w & 0 & 0 \\ 0 & 0 & 0 & I_r & 0 \\ 0 & 0 & 0 & 0 & M_w \end{bmatrix} \begin{Bmatrix} \ddot{x}_w \\ \ddot{\varphi}_w \\ \ddot{y}_w \\ \ddot{\theta}_w \\ \ddot{z}_w \end{Bmatrix} + \begin{bmatrix} c_t & -c_t d & 0 & 0 & 0 \\ -c_t d & c_r + c_t d^2 & 0 & -\Omega I_z & 0 \\ 0 & 0 & c_t & c_t d & 0 \\ 0 & \Omega I_z & c_t d & c_r + c_t d^2 & 0 \\ 0 & 0 & 0 & 0 & c_z \end{bmatrix} \begin{Bmatrix} \dot{x}_w \\ \dot{\varphi}_w \\ \dot{y}_w \\ \dot{\theta}_w \\ \dot{z}_w \end{Bmatrix} + \begin{bmatrix} k_t & -k_t d & 0 & 0 & 0 \\ -k_t d & k_r + k_t d^2 & 0 & 0 & 0 \\ 0 & 0 & k_t & k_t d & 0 \\ 0 & 0 & k_t d & k_r + k_t d^2 & 0 \\ 0 & 0 & 0 & 0 & k_t \end{bmatrix} \begin{Bmatrix} x_w \\ \varphi_w \\ y_w \\ \theta_w \\ z_w \end{Bmatrix} = \begin{Bmatrix} F_x \\ M_x \\ F_y \\ M_x \\ F_z \end{Bmatrix} \quad (1)$$

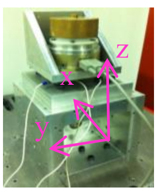
where x_w, y_w, z_w, φ_w, θ_w are RWA Degrees of Freedom (DoFs); F_x, F_y, F_z, M_x and M_y are excitations. Due to cantilevered configuration, motions in radial translational (x_w and y_w) and rotational DoFs (φ_w and θ_w) are coupled. This is the fundamental difference from mid-span configured RWAs. The axial translational motion is assumed as an independent mass-spring-damper system.

RWA Disturbance Test

A platform was designed to measure disturbances in hard-mounted boundary condition for the RWA, see Fig. 3 a). A detailed description of the platform is presented in [2]. The complete measurement of disturbances in the six DoFs requires three test setups. Example of F_z, M_x and M_y test setup for the microvibration testing is shown in Fig. 3 b). Note M_z is obtained from F_x and F_y test results.

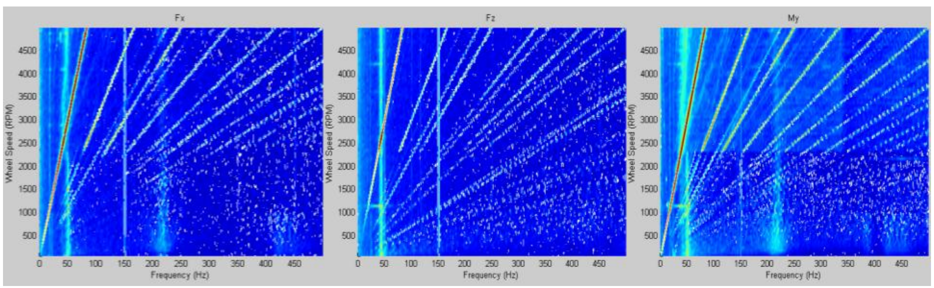


a) Platform



b) F_z, M_x and M_y Test

Figure 3. RWA test



a) F_x b) F_z c) M_y

Figure 4. Spectrum maps of RWA disturbance test results

In each test, RWA was spun from 60 to 6000 rpm with a step 60 rpm. Signals are sampled at 2048 Hz giving a frequency band up to 1024 Hz after FFT. 5s data were acquired at each speed. F_x, F_z, and M_y test result are plotted as spectrum maps and shown in Fig. 4. Due to axisymmetry, F_y and M_x disturbances are similar to F_x and M_y therefore only the first two are shown in the plot.

Typical dynamic features of a RWA such as harmonics and structural modes are identified from each figure. The radial translational mode at 50 Hz in Fig. 4 a), the axial translational mode at 44 Hz in Fig. 4 b) and the rocking mode starts from 26 Hz in Fig. 4 c). Strong nonlinearity was also seen, e.g. from Fig. 4 a) as speed increases up to about 2280 rpm, a clear amplitude jump is presented.

Sine-sweep (0.2 g input) tests were carried out to investigate this problem. In the test, forward and backward sweeps were performed and results were plotted as solid and dashed lines in Fig. 5. Accelerometer 3 and 4 were attached on top and side surface to measure in axial and radial DoFs.

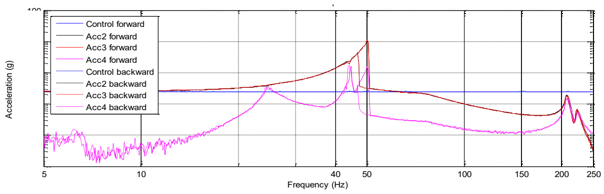
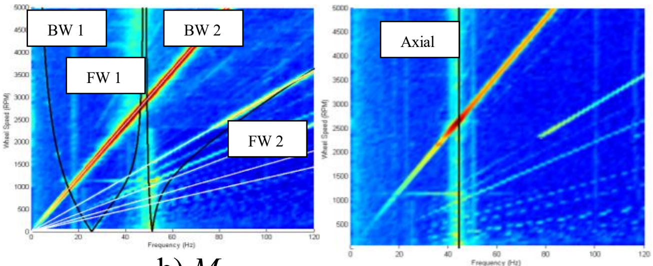


Figure 5. High level sine-sweep test results



b) M_y

c) F_z

Figure 6. Model and test results comparision

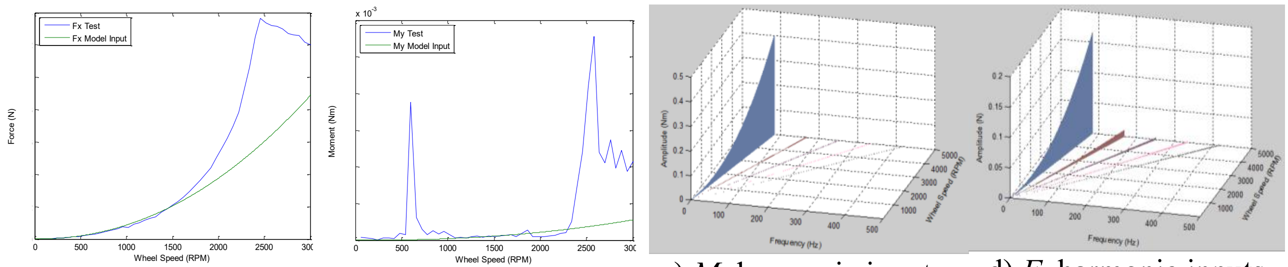
Fig. 5 has shown a classic nonlinear behaviour of a dynamic system. The only unmatched region is between 42 to 50 Hz, which is exactly the place the nonlinear amplitude jump occurred in disturbance tests. The extracted damping ratio with RWA finite element model is 0.18. Meanwhile, low level sine-sweeps (0.005 g input) have also been performed. In this case nonlinearity is not excited and the damping ratio is found to be 0.014. Fig. 6 shows structural modes calculated from the model and superimposed with test results in M_y and F_z. In M_y, all structural modes have been precisely captured by the model particularly at low speeds below 2280 rpm, but due to strong nonlinearity, the model is slightly different from test results in high speed region. This also causes slight difference in primary resonances between the two. In F_z however, the axial translational mode in both model and test results are constant. They are not influenced by nonlinearity as expected.

Harmonic Inputs

Input excitations comprise harmonic inputs and broadband noises. For each DoF, the total harmonic input is the superposition of all harmonics. In this paper, the first five integer harmonics are modelled. Model parameters such as amplitude coefficients are essential for harmonic input modeling. Amplitude coefficients are simply mass imbalances for fundamental harmonics, while for higher harmonics they must be extracted from test results. A new method is developed in this thesis for modeling harmonic inputs. The method is based on disturbance test results, i.e. a semi-empirical method. First of all, a program is written to find the “cut-off” speed. The “cut-off” speed is maximum speed of regions where primary resonances have not exerted their influences on fundamental harmonic responses. It is found in this case 1500 rpm. The rest of simulations can be understood as a trade-off between the speed power and amplitude coefficients until the simulated harmonic inputs match with the corresponding harmonic response test results up to the “cut-off” speed. Examples of simulates results are shown in Fig. 7.

Linear Harmonic Responses

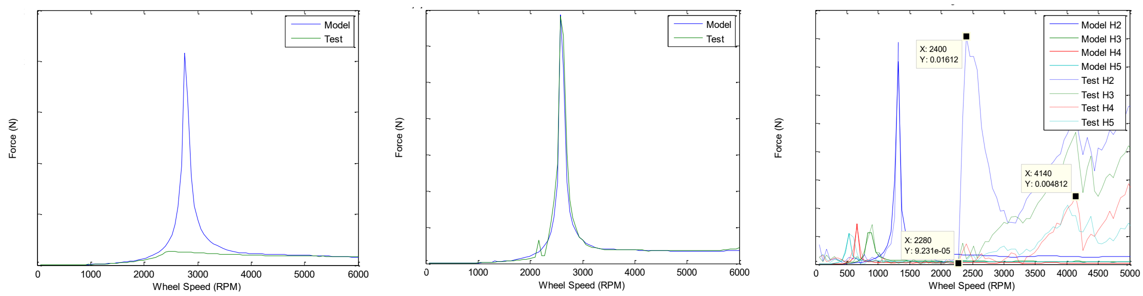
Linear harmonic response forces and moments are simulated using the state space approach. Harmonic responses are simulated for two cases based on suspension system damping properties.



a) F_x H1 input b) M_y H5 input c) M_y harmonic inputs d) F_z harmonic inputs

Figure 7. Simulated harmonic inputs

Damping values are adopted from low level sine-sweep tests, i.e. linear damping parameters. In practice, damping values vary as excitations change with test speeds. It is expected this model will not capture the highly damped resonances. Damping values need to be reduced and high level sine-sweep tests are performed. The simulated harmonic responses with linear damping are shown in Fig. 8.



a) F_x H1 model b) F_z H1 model c) F_z higher harmonics

Figure 8. Simulated harmonic responses compared with test results.

It was found that simulated fundamental harmonic (H1) responses in radial DoFs could not match the test results at resonances due to high damping of the suspension system. In axial translational DoF, the model matches very well with the test results, indicating the low damping is valid. Higher harmonics are also simulated for each DoF. However due to much lower higher harmonic inputs and lightly damped by the suspension system resulting in sharp peaks at resonances in all cases. Note after 2280 rpm, the nonlinear responses including resonances that not considered in the model are not simulated. Next, damping values from high level sine-sweep tests are used and the simulated F_x H1 responses are presented in Fig. 9. The model is better matched with test results.

Broadband Noise Model Development

The broadband noise model is developed based on empirical method by removing identified spikes, i.e. harmonics and resonances. Spikes are identified from cumulative RMS value plot. Since every distinctive spike also indicates a significant amount energy change and appears as a step in CRMS plot, see Fig. 10 a), thus can be used to identify spikes. A critical RMS value is needed to automatically find the spikes that fall within criteria. The critical RMS value is defined from background noises by calculating the RMS value of peaks at frequency band 20 Hz, the average is calculated. As energy increase smoothly, the difference between any two peaks in the band is calculated and the average value in the selected frequency band (200 to 500 Hz in this case) is used as the critical value to identify harmonics in response test results at each speed.

Identified harmonics are then removed using typical bandstop filters, filter parameters are chosen so it only removes identified peaks and have little influences on responses around them (cut-off frequency band ± 0.5 Hz). To guarantee the resulting amplitudes at removed spikes are close to general broadband noise level, an iterative removing process is performed and gradually reducing amplitudes of identified peaks until all removed. For example, the resulting broadband noise of F_z at 1800 rpm is shown in Fig. 10 b).

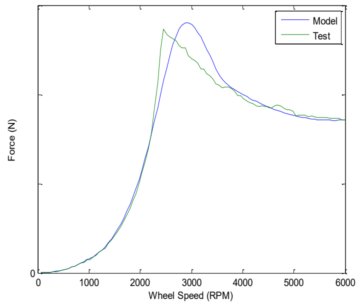
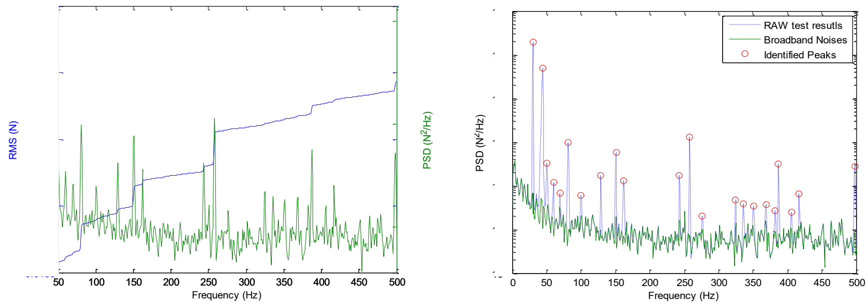


Figure 9. F_x H1 model (high level)

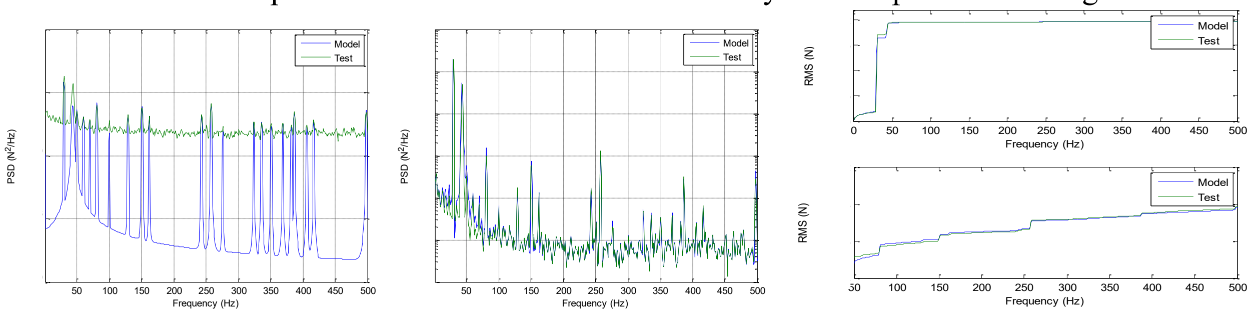


a) CRMS b) Broadband nosies

Figure 10. Spike identification and broadband noises (F_z 1800 rpm)

Full Model Validation

RWA full disturbance model includes excitations from both identified harmonics and broadband noises. Simulated responses under harmonic excitations only are first presented in Fig. 11.



a) Harmonic responses b) Full responses c) Full model RMS compared to test

Figure 11. RWA full disturbance simulation (F_z at 1800 rpm)

From Fig. 11 a), it is clear that simulated disturbance without considering broadband noise have flaws on the general responses apart from identified peaks, although they do not contribute much to the total energy in the signal. Fig. 11 b), on the other hand, has fully validated the disturbance model. The general response levels are levelled up, note they also slightly increase amplitudes of identified peaks, they can be corrected in simulating harmonic responses. Fig. 11 c) shows CMRS values of the full model, again the full model is very well matched with test results.

Summary

In this paper, disturbances induced by a cantilevered RWA have been fully simulated and validated against test results. The model takes account of identified harmonics and broadband noise excitations. Due to nonlinearity and great change in suspension system damping values in the test speed range, the model cannot precisely predict higher harmonic response in the radial DoFs.

Acknowledgments

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